

Lab Assignment 03

Cats Versus Dogs – Image Classification Using CNNs with Transfer Learning

1. Problem Statement

Develop an image classification system that can automatically classify an input image as either **Cat** or **Dog** using Convolutional Neural Networks (CNNs).

Instead of training CNNs entirely from scratch, students will use **transfer learning** with multiple pre-trained CNN architectures. The objective is to experimentally determine **which pre-trained model provides the best trade-off between predictive performance, training time, model size, and computational requirements** for the cats-versus-dogs classification problem.

2. Aim

To build, train, evaluate, and compare multiple transfer-learning-based CNN models for binary image classification of cats and dogs, and identify the best-performing model using appropriate evaluation criteria.

3. Objectives

Students should be able to:

1. Understand the concept of **transfer learning** in deep learning.
2. Prepare and preprocess an image dataset for CNN training.
3. Apply **data augmentation** to improve model generalization.
4. Use pre-trained CNN architectures for image classification.
5. Perform **feature extraction** using frozen convolutional layers.
6. Perform **fine-tuning** of selected layers of a pre-trained CNN.
7. Compare multiple transfer learning architectures experimentally.
8. Evaluate binary classifiers using multiple performance metrics.
9. Analyze training/validation curves and identify overfitting.
10. Select the best model based on **performance as well as computational efficiency**..

4. Dataset

Recommended Dataset

Students may use a publicly available Cats and Dogs image dataset given below.

<https://www.kaggle.com/datasets/samueltcortinhas/cats-and-dogs-image-classification>

The dataset contains two classes:

Class	Label
Cat	0
Dog	1

The reference project uses separate Cat and Dog directories and demonstrates loading image batches rather than requiring the entire image collection to be held in memory.

Directory Structure

Version 4 (69.26 MB)

- ▼ test
 - cats
 - dogs
- ▼ train
 - cats
 - dogs

Summary

- 697 files

5. Dataset Understanding

Students should investigate the dataset before beginning modeling.

Task 1: Dataset Exploration

Load the dataset and answer:

1. How many images are present?
2. How many images belong to each class?
3. Are the two classes balanced?
4. What are the minimum, maximum, and average image dimensions?
5. Are all images RGB?
6. Are there corrupted or unreadable images?

Task 2: Visual Exploration

1. Display a grid containing at least 12 randomly selected images.
2. The visualization should show:
3. Original image
4. Class label
5. Image dimensions, where appropriate

Expected Output

- 4X3 Grid containing 12 randomly selected images.

6. Image Preprocessing

CNNs require images to have a consistent input size.

Task 1: Image Resizing

Resize all images to: $128 \times 128 \times 3$

Students should explain:

- Why resizing is necessary
- Why RGB images have three channels
- What information may be lost during resizing

Task 2: Pixel Normalization

Normalize pixel values from $[0, 255]$ to $[0, 1]$

Students should explain why normalization is useful during neural network training.

Students must ensure that the **test data is not used during model training or hyperparameter tuning.**

7. Data Augmentation

CNNs can overfit when the training dataset is limited. Image augmentation can create additional variations of existing training images. The reference material specifically discusses rotation, translation, horizontal flipping, and zooming as augmentation strategies.

Task 1: Apply Data Augmentation

Implement suitable transformations such as:

- Random rotation
- Random horizontal flip
- Random zoom
- Random translation

Task 2: Visualization

Select one original image and generate **at least five augmented versions**.

Students Must Discuss

1. How does augmentation increase data diversity?
2. How can augmentation reduce overfitting?
3. Why should augmentation be applied only to training data?
4. What could happen if augmentation is too aggressive?

8. Transfer Learning Models to Compare

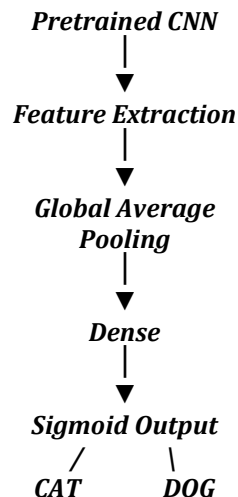
Task 1

Students **must implement at least five** pre-trained CNN architectures.

Recommended Models

Model	Type	Main characteristic
VGG16	Deep CNN	Simple and widely studied architecture
ResNet50	Residual CNN	Uses residual/skip connections
MobileNetV2	Lightweight CNN	Designed for computational efficiency
EfficientNetB0	Efficient CNN	Balances accuracy and computational cost
Xception	CNN	Uses depthwise separable convolutions

Suggested Architecture



Steps

Step 1 – Load the Pre-trained Model

Use ImageNet-pretrained weights.

Step 2 – Remove the Original Classification Head

Step 3 – Freeze the Base Network

Step 4 – Add a New Classification Head

9. Train the Feature-Extraction Model

Initially, the pre-trained CNN must remain frozen.
Only the newly added classification head should be trained.

Suggested Configuration

Parameter	Suggested value
Optimizer	Adam
Loss	Binary Cross-Entropy
Batch size	16/32
Epochs	10–20
Activation	ReLU
Output activation	Sigmoid
Early stopping	Yes

10. Fine-Tuning

After training the classifier head, students must investigate **fine-tuning**.

Procedure

Stage 1

- Freeze the complete pre-trained network.
- Pre-trained layers → Frozen
- Classification head → Trainable

Stage 2

- Unfreeze selected deeper layers.
- Early layers → Frozen
- Middle layers → Frozen
- Later layers → Trainable
- Classification head → Trainable

Use a **small learning rate** during fine-tuning.

11. Compare Five Transfer Learning Models

Students must conduct a controlled comparison and present their findings in tables.

Model	Parameters	Trainable Parameters	Test Accuracy	Precision	Recall	F1	AUC	Training Time
VGG16								
ResNet50								
MobileNetV2								
EfficientNetB0								
Xception								

12. Hyperparameter Tuning

For the **top two models**, perform systematic hyperparameter tuning.

Investigate at least **three** of the following:

- Learning rate
- Batch size
- Dropout
- Number of dense neurons
- Number of unfrozen layers
- Optimizer
- Data augmentation strength

Example:

Experiment	Learning Rate	Dropout	Unfrozen Layers	Validation Accuracy
1	1e-3	0.2	0	
2	1e-4	0.3	10	
3	1e-5	0.5	20	
4	1e-5	0.3	30	

Students should explain **why the selected configuration was finally chosen**.

13. Evaluation of the Final Models

For the best-performing models, generate:

Required

1. Confusion matrix
2. Classification report
3. ROC curve
4. AUC
5. Training vs validation accuracy
6. Training vs validation loss

14. Error Analysis

This is an **important research-oriented component** of the laboratory.

Identify:

A. Correct Predictions

Display examples of:

- Correctly classified cats
- Correctly classified dogs

B. Incorrect Predictions

Display examples of:

- Cat predicted as Dog
- Dog predicted as Cat

C. Difficult Images

Identify images where the model has low confidence and investigate possible reasons.

15 Best Model Selection

Students must **not select the best model based only on accuracy**.

The final model should be selected considering:

Predictive Performance + Generalization + Computational Efficiency

Consider:

Criterion	Importance
Test accuracy	High
F1-score	High
AUC	High
Generalization gap	High
Training time	Medium
Number of parameters	Medium
Model size	Medium
Inference time	Medium
Deployment suitability	Medium

Final Question

If two models have almost identical accuracy, which model should be selected and why?

Students must provide an evidence-based answer using their experimental results.

16. Deployment

Students must deploy the **best-performing transfer learning model** as a simple image-classification application.

Deployment Requirements

- i. **Save the final trained model** in an appropriate format such as .keras or .h5.
- ii. **Develop a user interface** that allows the user to upload a cat/dog image.
- iii. The application should:
 - Accept an image as input.
 - Display the uploaded image.
 - Perform the same preprocessing used during model training.
 - Load the saved trained model.
 - Predict whether the image is **Cat** or **Dog**.
 - Display the predicted class.
 - Display the prediction confidence/probability.
- iv. **Recommended deployment platforms:**
 - Streamlit — preferred
 - Gradio
 - Flask/FastAPI — optional advanced implementation

17. Report preparation

Students should prepare the lab report using the following sections:

1. Title
2. Aim
3. Problem Statement
4. Objectives
5. Dataset Description
6. Software and Hardware Requirements
7. Dataset Exploration and EDA
8. Image Preprocessing
9. Data Augmentation
10. Baseline CNN Model
11. Transfer Learning Methodology
12. Transfer Learning Models Used
 - VGG16
 - ResNet50
 - MobileNetV2
 - EfficientNetB0
 - Xception
13. Experimental Setup
14. Feature Extraction Results
15. Fine-Tuning Results
16. Hyperparameter Tuning
17. Performance Evaluation
18. Comparative Analysis of Models
19. Error Analysis
20. Best Model Selection and Justification
21. Conclusion
22. Future Scope
23. References
24. Appendix – Python Code

18. Final Deliverables

- i. Jupyter Notebook / Google Colab Notebook
- ii. Dataset or Dataset Access Link
- iii. Image Preprocessing and Data Augmentation Pipeline
- iv. EDA Visualizations
- v. Baseline CNN Implementation
- vi. Multiple Transfer Learning Model Implementations
- vii. Feature Extraction and Fine-Tuning Results
- viii. Hyperparameter Tuning Results
- ix. Model Evaluation Results and Visualizations
- x. Model Comparison Table

- xi. Final Trained Model
- xii. Deployment Application / Source Code
- xiii. Screenshots of Deployment
- xiv. Lab Report